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## 0.1 Cantor Set

Define operations on  $\mathcal{P}(\mathbb{R})$  as: for  $A, B \subset \mathbb{R}$  and  $c \in \mathbb{R}$ ,

$$A + B \stackrel{\text{def}}{=} \{a + b \mid a \in A, b \in B\}$$

$$cA \stackrel{\text{def}}{=} \{ca \mid a \in A\}$$

If  $A = \{\alpha\}$  is a singleton, denote  $A + B$  as  $\alpha + B$ .

**Definition 0.1.1.** Let  $C_0 \stackrel{\text{def}}{=} [0, 1]$  and  $C_n \stackrel{\text{def}}{=} \frac{C_{n-1}}{3} \cup \left(\frac{2}{3} + \frac{C_{n-1}}{3}\right)$ , for  $n \geq 1$ .

An intersection  $C = \bigcap_{n=1}^{\infty} C_n$  is called a *Cantor Set*.

Cantor set is a remarkable object in the Real numbers set.

In particular, Cantor Set is *Uncountable, Compact, Perfect, Nowhere Dense*, and has *Measure zero*.

Moreover, sum of two Cantor sets is  $[0, 2]$ .

And, total length (or measure) of  $C$  is zero, because:

$$\#C_0 = 1,$$

$$\#C_1 = \frac{2}{3} = 1 - \frac{1}{3},$$

$$\#C_2 = 1 - \frac{1}{3} - \frac{1}{3^2} \cdot 2,$$

⋮

$$\#C_n = 1 - \sum_{i=0}^n \frac{2^i}{3^{i+1}} = 1 - \frac{1}{3} \cdot \frac{1 - \left(\frac{2}{3}\right)^{n+1}}{1 - \frac{2}{3}} = 1 - \left(1 - \left(\frac{2}{3}\right)^{n+1}\right) = \left(\frac{2}{3}\right)^{n+1}.$$

## 0.2 Ternary Expansion

**Definition 0.2.1.** Given  $x \in (0, 1)$ , a sequence  $\{n_i\}_{i=1}^{\infty}$  in  $\{0, 1, 2\}$  is called a *ternary expansion* of  $x$  if:

$$x = \sum_{i=1}^{\infty} \frac{n_i}{3^i}$$

**Theorem 0.2.2.** Given  $x \in (0, 1)$ , a ternary expansion of  $x$  exists.

*Proof.* Let  $x \in (0, 1)$  be given. We construct the sequence  $\{n_i\}$  inductively as follows:  
First, choose  $n_1 \in \{0, 1, 2\}$  to be the largest integer such that

$$\frac{n_1}{3} < x$$

Next, for chosen  $n_1, n_2, \dots, n_{k-1}$ , we choose the largest digit  $n_k \in \{0, 1, 2\}$  such that

$$\frac{n_1}{3} + \frac{n_2}{3^2} + \dots + \frac{n_{k-1}}{3^{k-1}} + \frac{n_k}{3^k} < x$$

From this construction of  $\{n_i\}$ , it follows that for any  $k \in \mathbb{N}$ ,

$$\sum_{i=1}^k \frac{n_i}{3^i} < x \leq \sum_{i=1}^k \frac{n_i}{3^i} + \frac{1}{3^k}$$

Note that the partial sum  $\sum_{i=1}^k \frac{n_i}{3^i}$  increases monotonically and is bounded by 1, thus it converges.  
Taking the limit as  $k \rightarrow \infty$  in the above inequality, we obtain:

$$\sum_{i=1}^{\infty} \frac{n_i}{3^i} \leq x \leq \sum_{i=1}^{\infty} \frac{n_i}{3^i} + \lim_{k \rightarrow \infty} \frac{1}{3^k}$$

Since  $\lim_{k \rightarrow \infty} \frac{1}{3^k} = 0$ , we have

$$x = \sum_{i=1}^{\infty} \frac{n_i}{3^i}$$

which gives the desired result. □

Note: the ternary expansion described in the existence theorem is called the *First-form* ternary expansion.

**Theorem 0.2.3.** Given  $x \in (0, 1)$ , the ternary expansion of  $x$  is unique except: for some  $m \in \mathbb{N}$  and  $0 < a < 3^m$ ,

$$x = \frac{a}{3^m}, \text{ irreducible form} \quad (*)$$

Particularly, if  $x$  can be represented as  $(*)$ , it has only two different ternary expansion.

*Proof.* First, we show that if  $x$  is of the form  $(*)$ , it has only two expansions. If  $x$  can be represented as  $(*)$ , then

$$x = \frac{a}{3^m} = \frac{a_1}{3} + \frac{a_2}{3^2} + \cdots + \frac{a_m}{3^m}$$

where  $a_i \in \{0, 1, 2\}$  with  $a_m \neq 0$ . It is obtained by the Euclidean Algorithm:

$$a = 3^{m-1}a_1 + 3^{m-2}a_2 + \cdots + 3^2a_{m-2} + 3a_{m-1} + a_m$$

(If  $a_m = 0$ , then  $x$  can be divided by 3, it is contradiction to irreducibility.) Meanwhile, note that

$$\sum_{k=m+1}^{\infty} \frac{2}{3^k} = \frac{2}{3^{m+1}} \cdot \frac{1}{1 - 1/3} = \frac{2}{3^{m+1}} \cdot \frac{3}{2} = \frac{1}{3^m}.$$

Therefore, we obtain two ternary expansion of  $x$  as:

$$x = \frac{a_1}{3} + \frac{a_2}{3^2} + \cdots + \frac{a_m}{3^m} + \sum_{k=m+1}^{\infty} \frac{0}{3^k} = \frac{a_1}{3} + \frac{a_2}{3^2} + \cdots + \frac{a_m - 1}{3^m} + \sum_{k=m+1}^{\infty} \frac{2}{3^k} \quad (**)$$

This proves that such  $x$  has two distinct expansions. Meanwhile, suppose that  $x$  has two distinct expansions  $\{n_i\}, \{r_i\}$ . Let  $j \in \mathbb{N}$  be the smallest index such that  $n_j \neq r_j$ . Since  $n_i = r_i$  for all  $i < j$ , we have:

$$0 = \sum_{i=1}^{\infty} \frac{n_i - r_i}{3^i} = \frac{n_j - r_j}{3^j} + \sum_{i=j+1}^{\infty} \frac{n_i - r_i}{3^i} \quad (***)$$

Since  $n_j \neq r_j$  and

$$\left| \sum_{i=j+1}^{\infty} \frac{n_i - r_i}{3^i} \right| \leq \sum_{i=j+1}^{\infty} \frac{2}{3^i} = \frac{1}{3^j}$$

the  $(***)$  holds if and only if

$$n_j - r_j = 1, n_i - r_i = -2 \text{ for all } i \geq j+1 \text{ or } n_j - r_j = -1, n_i - r_i = 2 \text{ for all } i \geq j+1$$

But this equivalent to  $x$  can be represented in the form of  $(*)$ , therefore we proved the Uniqueness.  $\square$

Note: the ternary expansion is of the form  $(*)$  has two expansions. Clearly, the left side in  $(**)$  is a first form. The right side form is called a *Second form*.

### 0.3 Ternary representation of Cantor Set

**Theorem 0.3.1.**  $C = \left\{ \sum_{i=1}^{\infty} \frac{n_i}{3^i} \mid \{n_i\} \subseteq \{0, 2\} \right\}$  where  $C$  is the Cantor set.

*Proof.* Since the existence of ternary expansion,

$$C_0 = [0, 1] = \left\{ \sum_{i=1}^{\infty} \frac{n_i}{3^i} \mid \{n_i\} \subseteq \{0, 1, 2\} \right\}$$

Observe that

$$\begin{aligned} C_1 &= \frac{C_0}{3} \cup \left( \frac{2}{3} + \frac{C_0}{3} \right) = \left\{ \frac{0}{3^1} + \sum_{i=2}^{\infty} \frac{n_i}{3^i} \mid \{n_i\}_{i=2}^{\infty} \subseteq \{0, 1, 2\} \right\} \cup \left\{ \frac{2}{3^1} + \sum_{i=2}^{\infty} \frac{n_i}{3^i} \mid \{n_i\}_{i=2}^{\infty} \subseteq \{0, 1, 2\} \right\} \\ &\vdots \\ C_k &= \bigcup_{n_1, \dots, n_k \in \{0, 2\}} \left\{ \frac{n_1}{3^1} + \frac{n_2}{3^2} + \dots + \frac{n_k}{3^k} + \sum_{i=k+1}^{\infty} \frac{n_i}{3^i} \mid \{n_i\}_{i=k+1}^{\infty} \subseteq \{0, 1, 2\} \right\} \\ &= \left\{ \sum_{i=1}^k \frac{n_i}{3^i} + \sum_{i=k+1}^{\infty} \frac{n_i}{3^i} \mid \{n_i\}_{i=1}^k \subseteq \{0, 2\} \text{ and } \{n_i\}_{i=k+1}^{\infty} \subseteq \{0, 1, 2\} \right\} \end{aligned}$$

Finally,

$$C \stackrel{\text{def}}{=} \bigcap_{k \in \mathbb{N}} C_k = \left\{ \sum_{i=1}^{\infty} \frac{n_i}{3^i} \mid \{n_i\} \subseteq \{0, 2\} \right\}$$

□

**Theorem 0.3.2.**  $C + C = [0, 2]$  where  $C$  is the Cantor set.

*Proof.* Since  $C \subseteq [0, 1]$ ,  $C + C \subseteq [0, 2]$  is clear. Let  $x \in [0, 2]$  be given.

Trivially  $\frac{x}{2} \in [0, 1]$ . Therefore, there exists a ternary expansion  $\{n_i\} \subseteq \{0, 1, 2\}$  s.t.

$$\sum_{i=1}^{\infty} \frac{n_i}{3^i} = \frac{x}{2}$$

Use this fact, we can write that

$$x = \sum_{i=1}^{\infty} \frac{2n_i}{3^i}$$

Define

$$a_i = \begin{cases} 0 & \text{if } n_i = 0 \\ 0 & \text{if } n_i = 1 \\ 2 & \text{if } n_i = 2 \end{cases} \quad \text{and} \quad b_i = \begin{cases} 0 & \text{if } n_i = 0 \\ 2 & \text{if } n_i = 1 \\ 2 & \text{if } n_i = 2 \end{cases}$$

Then,  $2n_i = a_i + b_i$  for all  $i \geq 1$ . And,

$$x = \sum_{i=1}^{\infty} \frac{2n_i}{3^i} = \sum_{i=1}^{\infty} \frac{a_i}{3^i} + \sum_{i=1}^{\infty} \frac{b_i}{3^i} \in C + C$$

□

A

Prop. Let  $C$  be a Cantor set, and  $\chi_C$  be a characteristic function defined on  $[0,1]$

- 1). The map  $\chi_C$  is continuous on  $[0,1] \setminus C$ .
- 2). The map  $\chi_C$  is discontinuous at every point on  $C$ .
- 3). The map  $\chi_C$  is differentiable on  $[0,1] \setminus C$ .
- 4). The map  $\chi_C$  is Riemann-Integrable on  $[0,1]$ .

Proof. All  $C_n$  ( $n \in \mathbb{N}$ ) are closed, so  $C = \bigcap_{n \in \mathbb{N}} C_n$  is also closed set.

Therefore, given  $x \in [0,1] \setminus C$ , there exists  $r > 0$  such that  $(x-r, x+r) \subseteq [0,1] \setminus C$ .

Now, for all  $t \in (x-r, x+r)$ ,  $|\chi_C(x) - \chi_C(t)| = |0 - 0| = 0$ , so  $\chi_C$  is continuous.

Similarly, 3) can be proved. To show the second assertion, let  $x \in C$  be given

Fix  $\varepsilon = \frac{1}{2}$ . Given  $\delta > 0$ , choose  $n \in \mathbb{N}$  such that  $1/3^n < \delta$ . Denote  $B = (x-\delta, x+\delta) \cap [0,1]$ .

By definition,  $x \in C_n$  and there exists  $t \in B \setminus C_n$ , because:

If no such  $t$  exists, that is,  $B \setminus C_n = \emptyset$ , it equals to  $B \subseteq C_n$ .

But, the  $C_n$  is union of disjoint closed intervals of length  $1/3^n$ , it contradicts  $1/3^n < \delta$ .

Thus,  $C \subseteq C_n$  implies  $t \in B \setminus C_n \subseteq B \setminus C$ , it implies

$$|f(x) - f(t)| = |1 - 0| = 1 > \frac{1}{2} = \varepsilon.$$

Consequently,  $\chi_C$  is not continuous at  $x \in C$ .

To show the last assertion, observe that

$$0 \leq \int_0^1 \chi_C(x) dx \leq \int_0^1 \chi_{C_n}(x) dx \leq \left(\frac{2}{3}\right)^n \cdot 3, \text{ because:}$$

Let  $P_n := \{a/3^n \mid a \in \mathbb{Z}, 0 \leq a \leq 3^n\}$  be a partition on  $[0,1]$ , then the upper sum

$$\begin{aligned} U(f, P_n) &= \sum_{i=1}^{3^n} M_i \cdot \frac{1}{3^n} \quad \text{where } M_i = \sup \left\{ \chi_C(x) \mid \frac{i-1}{3^n} \leq x \leq \frac{i}{3^n} \right\} \\ &\leq \frac{2^n \cdot 3}{3^n} \end{aligned}$$

so,  $\int_0^1 \chi_C \leq U(f, P_n)$  by the definition, we obtain the inequality.

Consequently, letting  $n \rightarrow \infty$ , we obtain the result.  $\square$

Prop. Let  $C$  be the Cantor set, and let  $f$  be a continuous at except  $C$ .  
Then,  $f$  is Riemann-integrable.